MD RAFI ISLAM

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Available from January 2027 (PhD defense and completion expected December 2026)

RESEARCH SUMMARY

PhD candidate in Electrical Engineering at The University of Alabama working on multimodal wearable sensing, embedded systems, signal processing, and machine learning for objective, real-world measurement of human movement, energy expenditure, and health behaviors. Co-designed the STM32-based Smart Lacelock foot-worn platform and first-authored a deep-learning study detecting sit-to-stand transitions and measuring their duration under nested participant-independent validation. Currently leads a calorimetry-referenced study estimating metabolic equivalents from foot-worn inertial and shoelace-tension signals under leave-one-subject-out validation with participant-level leakage control. Co-developed the embedded firmware and the open-source data-collection software for the FRIENDS platform, which objectively measures vaping behavior.

RESEARCH INTERESTS

Multimodal wearable sensing and sensor fusion | Embedded systems for physiological and behavioral measurement | Free-living energy expenditure estimation and indirect calorimetry reference methods | Human movement, gait, and mobility assessment | Machine learning for health behavior measurement, including representation learning under participant-independent validation | Intent detection for lower-limb prostheses and wearable robotics | Objective measurement of tobacco and nicotine product use | Reproducible and open-source research software

EDUCATION

PhD, Electrical Engineering | The University of Alabama, Tuscaloosa, AL | GPA 3.92 | Expected December 2026

Dissertation topic: Sensor-Based Human Movement Analysis

Advisor: Professor Edward Sazonov, Department of Electrical and Computer Engineering, The University of Alabama

MS, Electrical Engineering | The University of Alabama, Tuscaloosa, AL | GPA 3.92 | May 2026

BSc, Electrical and Electronic Engineering | Chittagong University of Engineering and Technology, Chittagong, Bangladesh | GPA 3.63 | November 2017

RESEARCH AND ACADEMIC APPOINTMENTS

Graduate Research Assistant | The University of Alabama | Tuscaloosa, AL | January 2021 – Present

Department of Electrical and Computer Engineering. Research across NSF- and NIH-funded wearable sensing programs spanning embedded hardware and firmware design, human-subject data collection, signal processing, and machine learning.

Lecturer | Premier University | Chittagong, Bangladesh | March 2018 – December 2020

Department of Electrical and Electronic Engineering. Taught Digital Signal Processing and Control Systems.

PUBLICATIONS

Full list: Google Scholar ID sG1AczMAAAAJ

Totals: 5 peer-reviewed journal papers, 5 peer-reviewed conference papers, 1 book chapter, 1 journal manuscript under review, 1 preprint.

Peer-Reviewed Journal Papers

1. Pranto SI, Fassler B, Islam MR, Schenkel A, Hawk LW, Sazonov E. FRIENDS GUI: A graphical user interface for data collection and visualization of vaping behavior from a passive vaping monitor. Journal of Open Research Software, volume 14, issue 1, article 24, 2026. DOI 10.5334/jors.613.
2. Islam MR, Shen X, Sazonov E. Sensors and Machine Learning Methods for Intent Detection in Lower-Limb Prosthetic Devices: A Review. IEEE Sensors Journal, volume 25, issue 12, pages 21054 to 21066, 2025. DOI 10.1109/JSEN.2025.3565195. (First author.)
3. Haque MR, Islam MR, Sazonov E, Shen X. Swing-phase detection of locomotive mode transitions for smooth multi-functional robotic lower-limb prosthesis control. Frontiers in Robotics and AI, volume 11, 2024. DOI 10.3389/frobt.2024.1267072.
4. Islam MR, Haque MR, Imtiaz MH, Shen X, Sazonov E. Vision-Based Recognition of Human Motion Intent during Staircase Approaching. Sensors, volume 23, 2023. DOI 10.3390/s23115355. (First author.)
5. Haque MR, Islam MR, Bassiri Z, Martelli D, Sazonov E, Shen X. Smart Lacelock: A Novel Shoelace Tensioning Device for Human Motion Sensing. IEEE Sensors Journal, volume 22, pages 18349 to 18358, 2022. DOI 10.1109/JSEN.2022.3197890.

Under Review and Preprints

1. Sazonov E, Fassler B, Islam MR, Schenkel A, Hossain D, Tiffany ST, Hawk LW. Vaping with FRIENDS: Flexible Robust Instrumentation of Electronic Nicotine Delivery Systems. IEEE Sensors Journal, under review, 2026.
2. Islam MR, Haque MR, Choma E, Hayes S, McMahon S, Shen X, Sazonov E. Sit-to-Stand Transitions Detection and Duration Measurement Using Smart Lacelock Sensor. arXiv preprint arXiv:2604.00175, 2026. (First author.)

Peer-Reviewed Conference Papers

1. Haque MR, Islam MR, Choma E, Hayes S, McMahon S, Sazonov E, Shen X. Smart Lacelock Sensor for the Balance Assessment of Community-Dwelling Older People. 45th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBC 2023. DOI 10.1109/EMBC40787.2023.10340165.
2. Islam A, Himu EA, Amin NB, Kaoser MS, Abrar F, Islam MR. Pulmonary Artery Pressure Prediction Based On Three Physiological And Seven ECG Signal Features. IEEE 3rd Global Conference for Advancement in Technology, GCAT 2022, pages 1 to 5. DOI 10.1109/GCAT55367.2022.9972058.
3. Islam MR, Pavel SR, Tunaz SA. Neurodegenerative Disease Classification Using Gait Signal Features and Random Forest Classifier. International Conference on Electrical Information and Communication Technology, EICT 2019. DOI 10.1109/EICT48899.2019.9068822. (First author.)
4. Pavel SR, Islam MR, Siddiqee AM. Fetal arrhythmia detection using fetal ECG signal. International Conference on Telecommunications and Photonics, ICTP 2019. DOI 10.1109/ICTP48844.2019.9041789.
5. Bhattacharjee A, Islam MR, Ullah MA. Weather Independent Positioning of MS Using Power Slope Patterning Algorithm via One BS. International Conference on Electrical, Computer and Communication Engineering, ECACE 2019. DOI 10.1109/ECACE.2019.8678919.

Book Chapter

1. Islam MR, Li S. Metaverse Security: Principles and Experiments. In Metaverse Platform Implementation, pages 131 to 149. CRC Press, 2025. DOI 10.1201/9781032663609. (First author.)

RESEARCH EXPERIENCE

Foot-Worn Multimodal Sensing for Energy Expenditure Estimation | NSF Grant No. 1734501 | 2021 – Ongoing

Role: project lead.

- Leads the development and validation of a foot-worn multimodal system (bilateral 9-DOF IMU plus shoelace-tension load cell, 512 Hz) for metabolic-equivalent (MET) estimation in 12 adults.
- Referenced the system breath-by-breath against gold-standard Cosmed K5 indirect calorimetry over a 14-activity laboratory protocol spanning sedentary to vigorous intensity.
- Develops three modeling arms in Python and PyTorch under a common leave-one-subject-out protocol on identical folds.
- Develops the feature-based arm: TSFEL statistical, temporal, and spectral features; three feature-selector arms and the feature count chosen in a nested inner leave-one-subject-out loop; ordinary-least-squares regression head; compared with a pre-specified nine-feature handcrafted model.
- Trains the reference arm, a two-foot dilated residual convolutional network, end to end with a squared-error loss.
- Pretrains the contrastive arm's encoder with a rank-based contrastive objective (Rank-N-Contrast), which orders embedding distances by label distance without discretizing MET, followed by a linear readout on the frozen embedding.
- Refits every fitted stage (input scaling, imputation, near-zero-variance and rank reduction, standardization, correlation pruning, feature selection) inside each leave-one-subject-out fold on training participants only, enforcing strict data-leakage control.
- Selects network width and training length in a nested inner loop that never sees the held-out participant.
- The reference convolutional arm (average of three training seeds) estimated bout-level MET in unseen participants over 14 activities spanning 0.8–9.8 MET with a per-participant mean absolute error of 0.614 MET (mean absolute percentage error 18.3 percent) and Lin's concordance correlation coefficient 0.904.
- Bland-Altman bias was -0.071 MET (95 percent limits of agreement -1.68 to +1.54 MET), with 71.8 percent agreement in intensity category (linearly weighted Cohen's kappa 0.71).
- The rank-based contrastive arm did not improve on it (matched per-participant difference +0.073 MET, 95 percent confidence interval -0.009 to +0.159), reported as a null result after a pre-specified convergence diagnostic. Manuscripts in preparation.
- Reports bout-level rather than window-level metrics with participant-clustered uncertainty, aligning inference with the effective sample size, and stratifies error by MET band and by participant.
- Benchmarks against non-sensor reference models (activity-class and demographic-only) to isolate the sensor's contribution to between-participant variation in the energy cost of the same activity.

Smart Lacelock Wearable Platform: Sit-to-Stand and Mobility Assessment | NSF Grant No. 1734501 | 2021 – Ongoing

Role: co-designer, first author on the sit-to-stand study. Sit-to-stand study code: github.com/mdrafiislam/smart_lacelock_sist.

- Co-designed the STM32L476-based embedded hardware and firmware stack for a shoe-mounted load-cell plus 9-DOF IMU

wearable (magnetometer not recorded) operating at 512 Hz, implementing interrupt- and DMA-driven acquisition, synchronized timestamping, and real-time BLE streaming.

- Built end-to-end MATLAB signal-processing pipelines covering filtering, normalization, and multi-resolution feature extraction for synchronized IMU and load-cell data in the sit-to-stand and balance-assessment studies.
- First-authored a deep-learning pipeline for sit-to-stand detection and duration measurement from shoe-mounted shoelace-tension and inertial signals in 16 community-dwelling older adults (3 men, 13 women; 71–83 years; mean 76.84, standard deviation 3.45), against a video-audited ground truth of 110 transitions.
- Built and compared five models under leave-one-participant-out validation across 16 folds: a 1D CNN (40,587 parameters), a CNN-BiLSTM (53,003 parameters), a 1D U-Net segmentation network (245,027 parameters), and two late-fusion ensembles.
- Reached a pooled event F1 of 0.787 with the selected configuration (95 percent confidence interval 0.708–0.853; precision 0.82, recall 0.75), with all five models between 0.711 and 0.787.
- Selected the training epoch, decision threshold, ensemble rule, stopping criterion, and model configuration inside every fold on three held-in inner-validation participants only, never on the held-out participant, with assertions verifying the three sets were disjoint.
- Showed that the evaluation grid alone moves an unchanged model's event F1 by 0.077, more than the 0.035 separating the four leading models, so window-level F1 cannot rank such configurations.
- Searched a 139-cell grid of window length and training stride, scoring every cell at every evaluation stride.
- Built a feature-based reference pipeline on identical folds and event rules: 2,856 features per window from the TSFEL catalogue plus bilateral-synchrony and load-cell morphology features, mRMR selection, and five learner families searched by seeded random search.
- Reached an event F1 of 0.695 with the feature-based pipeline, below every deep model.
- Quantified the shoelace-tension contribution by ablation: removing the load-cell channels lowered each individual network's event F1 by 0.051 to 0.187.
- Compared models on per-participant event F1 with a Friedman test (chi-squared(4) = 9.88, p = 0.043, Kendall's W = 0.15), Holm-corrected pairwise Wilcoxon signed-rank tests, rank-biserial effect sizes, and participant-bootstrap 95 percent confidence intervals over 10,000 draws.
- Reported that only one of ten pairwise comparisons survived correction, stating explicitly that a non-significant difference is not evidence of equivalence.
- Reported the post-hoc grid maximum (0.820) beside the inner-validation-selected configuration rather than in place of it, quantifying the selection bias at 0.028–0.073 event F1.
- Audited the annotation against synchronized 60 frame-per-second video at 0.1 s resolution, identified and corrected a per-participant video-sensor clock offset of 0.35–5.31 s, and adjudicated 30 additional stands outside the chair-stand block to build the 110-event label set.

FRIENDS Wearable Monitor for Objective Measurement of Vaping Behavior | NIH/NCI Award No. R01CA266170 | 2024 – Ongoing

Role: co-developer of firmware, data software, and device characterization.

- Co-developed bare-metal MSP430G2433 firmware for SPI flash logging, UART protocols, and real-time electromagnetic, capacitive-touch, and NTC thermistor sensing, passively recording puff and device-handling events together with mouthpiece aerosol temperature to onboard flash memory as 16-character hexadecimal event records.
- Implemented hardware-timer-based 64-bit extended POSIX timestamping at 1/32768 s (about 30.5 microsecond) resolution, with low-power modes supporting continuous multi-day (up to 10-day) free-living data collection.
- Executed schematic updates, PCB modifications, sensor calibration, and mixed-signal debugging for hardware bring-up.
- Co-designed the device enclosure in Autodesk Fusion 360 and produced prototype parts on an Afinia H800 fused-filament-fabrication 3D printer.
- Co-developed and validated the open-source Python FRIENDS GUI for binary-log decoding, automated puff and touch event detection, and structured dataset export.
- Validated the GUI to capture 99.46 percent of actual puff duration across 72 puffing events.
- Tested and documented 70-plus electronic nicotine delivery system devices using the VM-FRIENDS electromechanical vaping machine.
- Mapped electromagnetic detection-sensitivity zones, determined thermistor-assisted detection requirements, and documented deployment configurations.
- Built and documented the resulting compatibility library, which now covers 90-plus devices and directly informs the puff-detection algorithms used in the FRIENDS GUI software.
- Designed and executed fault-injection testing of the data pipeline: built a corrupted counterpart to a known-good log file introducing truncated timestamps, missing log lines, stripped prefix bytes, and malformed indentation.
- Verified through automated file comparison that every injected fault was detected and reported.

- Validation surfaced 17 electromagnetic false positives across the 24-hour protocol, removed by a 1.0 s minimum-duration filter, and confirmed timestamp-conversion drift below 10 s over a full day.
- Re-runs the device validation suite after firmware changes to detect regressions before deployment.

FRIENDS-2 STM32WB5 Sensor Board Firmware: Dual-Core Embedded Software, Storage, and USB | NIH/NCI Award No. R01CA266170 | January 2026 – Present

Role: lead developer, with Claude Code as AI pair programmer. Repository: github.com/mdrafiislam/FRIENDS_2_STM32WB5. Migrates the FRIENDS vaping monitor to the STM32WB5 processor so that sensor data can be streamed rather than only logged to onboard memory.

- Developed the complete bare-metal C application firmware for a custom sensor board built on the STM32WB5MMG dual-core wireless module, splitting work across the Cortex-M4 application core and the Cortex-M0+ BLE core.
- Implemented dual LIS2MDL magnetometer acquisition over SPI at 20 Hz, computing per-sensor magnitude and a differential vector that cancels the Earth's field, with boot-time baseline zeroing.
- Implemented event detection using TIM2 in PWM-input mode, measuring period and duty cycle of an external signal inside the interrupt service routine with running-mean hysteresis detection.
- Kept interrupt latency bounded: the ISR sets flags only, deferring SD card and USB work to the main loop.
- Built SD card logging on FatFs over SPI with a custom disk I/O layer, paired CSV run files, periodic sync, and safe close on stop.
- Implemented runtime USB class switching: the device boots as a CDC virtual COM port for commands and streaming, switches to USB Mass Storage on command so the host can read the SD card directly, and returns to CDC on eject.
- Implemented BLE streaming through the STM32_WPAN stack running on CPU2, with GATT notifications of sensor samples at 20 Hz and ATT MTU exchange on connect to carry full-size packets.
- Implemented software-triggered USB DFU: a serial command closes open files, detaches USB, sets a flag in an RTC backup register and resets, so the next boot jumps to the ST system bootloader before any clock or peripheral initialisation runs.
- Diagnosed and fixed a dual-core clock ownership fault by claiming the USB 48 MHz clock through a hardware semaphore, preventing the BLE core from powering it down, and added a runtime guard.

BLE Firmware and Low-Power Optimization for Wearable Sensors | NSF Grant No. 1734501 | 2023 – 2024

Role: firmware developer. Repository: github.com/mdrafiislam/BLE_nRF52832.

- Programmed the nRF52832 for high-throughput BLE communication at 8192 bit/s, optimizing GATT and GAP performance and DFU handling.
- Reduced current consumption from 1.6 mA to 180 microamps during transmission, and from 50 microamps to 6 microamps in advertising mode, extending device runtime for continuous free-living data collection.
- Encountered and worked within the RAM limits of the nRF52832 during BLE firmware development.
- Developed MATLAB-based real-time BLE acquisition and visualization tools.

Vision-Based Human Motion Intent Recognition for Assistive Devices | NSF Grant No. 1734501 | 2022 – 2023

Role: first author and lead developer.

- First-authored a YOLOv5 plus gradient-boosting intent-recognition system that identifies stair engagement versus avoidance from egocentric video, achieving 97.69 percent recognition at least two steps before the walking-to-stair-climbing mode transition.
- Created and annotated egocentric datasets comprising 12,187 staircase images and 375 approach videos across 25 staircases, enabling user-independent intent recognition for wearable robots and lower-limb prostheses.

TEACHING EXPERIENCE

Lecturer | Premier University, Chittagong, Bangladesh | March 2018 – December 2020

- Taught Digital Signal Processing, Department of Electrical and Electronic Engineering.
- Taught Control Systems, Department of Electrical and Electronic Engineering.
- Supervised two undergraduate student groups' final thesis projects.

Mentoring, Training, and Code Review | The University of Alabama | 2021 – Present

- Reviews laboratory colleagues' code, both manually and using Claude Code and Codex.
- Taught surface-mount soldering to about 7 undergraduates on Capstone Design teams (ECE 492 and ECE 494) in Fall 2025 and Spring 2026, and helped them debug microcontroller code and circuit boards.
- Trained two undergraduate students in vape device lifecycle testing.
- Currently working with a third undergraduate student on Bluetooth Low Energy integration for the AIM wearable food intake monitor.
- Trained the clinicians and clinical researchers who collected the data in the University of Minnesota sit-to-stand study and the FRIENDS study: what the sensor does, how to put it on the participants, and how to start, stop, and collect data with the co-created custom GUI.

PEER REVIEW AND PROFESSIONAL SERVICE

Reviewer for IEEE journals and conferences, 6 manuscripts reviewed:

- IEEE Engineering in Medicine and Biology Society, EMBS: 4 manuscripts reviewed
- IEEE Journal of Biomedical and Health Informatics, JBHI: 1 manuscript reviewed
- IEEE Sensors Journal: 1 manuscript reviewed

RESEARCH SOFTWARE AND DATA RESOURCES

FRIENDS GUI | github.com/eszsonic/FRIENDSGUI

Open-source Python GUI for data collection and visualization of vaping behavior from a passive ENDS monitor, deployed in ongoing ENDS and vaping-behavior research at The University of Alabama and the University at Buffalo, supported by NIH/NCI Award R01CA266170.

- Contributed software testing, validation, and maintenance: verified timestamp conversion, event decoding, data extraction, and consistency between the GUI executable and standalone script outputs.
- Co-authored, reviewed, and edited the software metapaper, published in the Journal of Open Research Software, volume 14, 2026, DOI 10.5334/jors.613; archived on Zenodo, DOI 10.5281/zenodo.18753292, under the MIT License.
- Validated to capture 99.46 percent of actual puff duration across 72 puffing events.

FRIENDS Vape Library | claws.eng.ua.edu/research-projects/44-friends/299-friends-vape-library

Open-source ENDS-device compatibility database covering 90-plus commercial devices, documenting electromagnetic detection-sensitivity zones, thermistor-assisted detection requirements, and deployment configurations for each device under test.

- Built and documented the library, which directly informs the puff-detection algorithms used in the FRIENDS GUI software, supporting tobacco and nicotine regulatory and public-health research.

RESEARCH FUNDING CONTEXT

Research conducted under the following awards. Role is research personnel; no principal investigator or co-investigator role is claimed.

- NSF Grant No. 1734501 – Smart Lacelock wearable platform, BLE firmware and low-power optimization, vision-based intent recognition.
- NIH/NCI Award No. R01CA266170 – FRIENDS wearable vaping behavior monitor and the FRIENDS-2 STM32WB5 sensor board firmware.

TECHNICAL SKILLS

Wearable and Physiological Sensing: Inertial measurement units (accelerometer, gyroscope), load cells, thermistors, capacitive touch sensing, electromagnetic sensing, ECG signal processing (conference papers, 2019 and 2022), multimodal sensor fusion, indirect calorimetry reference instrumentation (Cosmed K5)

Signal Processing: Filter design, z-normalization, sliding-window segmentation, temporal, statistical, and frequency-domain feature extraction, multi-resolution feature extraction, Kaiser-window FIR anti-alias filtering, zero-phase decimation, zero-phase Butterworth high-pass filtering, timestamp synchronization across sensor streams

Deep Learning: CNN-BiLSTM, 1D-CNN encoders, MLP regression heads, supervised contrastive representation learning (NT-Xent loss, cluster-based negative selection), rank-based contrastive learning for regression (Rank-N-Contrast), dilated residual 1D CNNs, 1D U-Net time-series segmentation, late-fusion model ensembling, YOLOv5 object detection

Classical Machine Learning: gradient boosting (XGBoost, LightGBM), AdaBoost, random forest, extra-trees, bagged and boosted trees, elastic-net logistic regression, support vector machines, k-nearest neighbors, regression for energy expenditure estimation

Validation Design: participant-independent cross-validation (leave-one-subject-out, k-fold), nested cross-validation with inner-validation model selection, data-leakage control through per-fold refitting, bout-level versus window-level metric reporting, participant-clustered uncertainty estimation, benchmarking against non-sensor reference models, event-level evaluation with IoU-based one-to-one matching, selection-bias quantification against post-hoc grid maxima

Statistics and Agreement Metrics: Friedman and Holm-corrected Wilcoxon signed-rank comparison, rank-biserial effect size, participant-bootstrap confidence intervals, evaluation metrics (accuracy, F1, precision, recall, RMSE, MAE, mean absolute percentage error), Bland-Altman bias and limits of agreement, Lin's concordance correlation coefficient, linearly weighted Cohen's kappa

Feature Engineering and Training: mRMR feature selection, grid-search hyperparameter tuning, class-imbalance handling (TSAUG, SMOTE), AdamW, class-weighted cross-entropy, time-series augmentation (jitter, gain, time-warp, channel masking), TSFEL feature catalogue, exploratory analysis of raw sensor data, result figures produced in matplotlib

Health and Behavior Analytics: Human movement and gait analysis, sit-to-stand transition detection, activity-specific energy expenditure estimation, puffing topography (puff duration, count, timing)

Human-Subject Research: IRB-approved protocol execution (calorimetry-referenced energy expenditure study), video-referenced

ground-truth annotation and annotation auditing, dataset curation and labeling

Embedded Firmware: Bare-metal C, C++, interrupt-driven architecture, DMA-based data transfer, low-power design, timing-critical control, 64-bit extended POSIX timestamping, SPI flash logging, binary log format design, BLE DFU firmware update handling

Microcontrollers and Platforms: STM32 (Cortex-M4, specifically STM32L476), MSP430 (specifically MSP430G2433), nRF52832, ESP32

Real-Time Operating Systems: FreeRTOS (basic), Zephyr (familiar)

Hardware Interfaces and Protocols: SPI (master and slave), I2C, UART, ADC, GPIO, Bluetooth Low Energy (GATT, GAP, DFU)

Hardware Bring-Up, Debug, and Validation: Oscilloscope-based signal inspection, JTAG and SWD debugging, hand soldering and board rework, PCB modification, schematic revision, sensor calibration, mixed-signal board bring-up

Mechanical Prototyping and Enclosure Design: Co-design of wearable device enclosures in Autodesk Fusion 360, additive manufacturing on an Afinia H800 fused-filament-fabrication 3D printer, iterative fit and form prototyping for body-worn hardware

Device Characterization and Test: Multi-device characterization campaigns, electromechanical test fixtures, unattended multi-day device logging, regression validation after firmware changes, binary log decoding and verification, executable-versus-script output consistency checking

Agentic and AI-Assisted Development: Claude Code (daily practice), Codex (daily practice), agent-assisted code review of colleagues' work, manual code review

Open-Source and Reproducibility: Python GUI development and validation, MIT-licensed software release, Zenodo archiving, GitHub maintenance and issue support, software metapaper authorship, pytest test suite for the sit-to-stand pipeline (github.com/mdrafiislam/smart_lacelock_sist)

Programming Languages: C, C++, Python, MATLAB

Machine Learning Frameworks and Scientific Python: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, SciPy, pandas, matplotlib

Development Tools: Git, GitHub, STM32CubeIDE, Code Composer Studio, Segger Embedded Studio, EasyPC Schematic and PCB Design, Autodesk Fusion 360, PyTorch

AWARDS AND HONORS

- Poster Presentation Competition, 2nd place, The University of Alabama, 2023 and 2024
- Graduate Research Assistantship with tuition support, The University of Alabama, 2021 – Present